

BAYAN MOD, China

WMO: 524950

Lat: 40.750N

Lon: 104.500E

Elev: 1329

StdP: 86.35

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.1	-20.5	-32.3	0.2	-13.9	-30.0	0.3	-13.2	11.3	-3.1	9.9	-5.1	0.9	40	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	33.9	16.0	32.6	15.4	31.2	15.0	19.2	27.7	18.1	26.7	17.3	25.7	4.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	14.0	21.8	15.6	13.0	21.2	14.4	12.0	21.1	61.0	27.9	57.1	26.6	54.1	25.5	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.9	10.1	8.8	DB	-26.8	36.8	3.0	1.8	-28.9	38.1	-30.7	39.2	-32.4	40.2	-34.5	41.5
			WB	-27.0	20.2	3.0	1.2	-29.1	21.1	-30.9	21.8	-32.5	22.5	-34.7	23.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	-10.4	-5.7	1.8	10.3	16.7	22.2	24.6	22.6	16.7	8.5	-0.6	-8.3	(6)
(7)		DBStd	12.83	4.44	5.01	5.37	4.91	4.09	3.15	2.86	3.06	3.74	4.46	4.74	4.52	(7)
(8)		HDD10.0	2354	632	440	256	54	3	0	0	2	81	319	568	(8)	
(9)		HDD18.3	4178	890	673	511	242	79	6	0	4	73	306	569	826	(9)
(10)		CDD10.0	1728	0	0	3	64	212	367	453	392	203	34	0	0	(10)
(11)		CDD18.3	511	0	0	0	3	29	122	195	137	24	1	0	0	(11)
(12)		CDH23.3	5567	0	0	0	53	402	1366	2144	1352	243	8	0	0	(12)
(13)		CDH26.7	2190	0	0	0	8	104	532	982	512	52	0	0	0	(13)
(14)	Wind	WSAvg	3.5	2.7	3.2	3.9	4.3	4.4	4.0	3.7	3.5	3.2	3.1	3.3	2.9	(14)
(15)	Precipitation	PrecAvg	89	1	0	2	3	7	12	25	25	9	4	1	0	(15)
(16)		PrecMax	199	5	2	10	15	30	44	87	71	30	20	18	6	(16)
(17)		PrecMin	38	0	0	0	0	0	0	4	0	0	0	0	0	(17)
(18)		PrecStd	39	1	1	2	4	8	12	19	18	8	5	4	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	10.9	20.1	27.8	31.4	34.7	37.0	34.6	31.4	24.6	15.2	6.6	(19)
(20)		MCWB	-1.3	1.7	6.8	10.6	12.6	14.5	18.2	17.1	14.5	10.4	4.6	-0.6	(20)	
(21)		2%	DB	2.3	7.7	16.4	24.6	29.3	32.8	34.6	32.7	27.9	21.4	12.5	3.8	(21)
(22)		MCWB	-3.2	0.0	4.8	9.0	11.8	14.3	16.4	15.9	13.2	9.0	3.4	-1.9	(22)	
(23)		5%	DB	-0.2	5.6	13.9	22.1	27.6	31.4	33.2	31.2	26.0	19.3	10.1	1.5	(23)
(24)		MCWB	-4.7	-1.1	3.5	7.9	11.1	13.8	16.1	15.4	12.6	8.1	2.1	-3.3	(24)	
(25)		10%	DB	-2.4	3.3	11.4	19.8	25.6	29.9	31.7	29.7	24.2	17.0	7.7	-0.6	(25)
(26)		MCWB	-6.1	-2.4	2.2	7.1	10.1	13.5	15.8	15.3	12.1	7.0	1.1	-4.6	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.2	2.4	7.2	11.6	14.5	18.1	20.7	20.6	17.6	11.9	5.4	-0.2	(27)
(28)		MDCB	5.2	10.0	18.8	23.0	26.3	26.4	32.2	26.9	24.3	20.6	12.9	6.2	(28)	
(29)		2%	WB	-3.0	0.3	5.2	10.1	13.0	16.8	19.4	19.3	15.9	10.2	3.9	-1.8	(29)
(30)		MDCB	2.0	6.8	15.1	21.6	25.3	25.0	29.2	26.5	23.2	18.9	11.1	3.7	(30)	
(31)		5%	WB	-4.5	-0.9	3.9	8.8	11.9	15.8	18.3	18.2	14.7	8.9	2.6	-3.2	(31)
(32)		MDCB	-0.3	5.1	12.5	20.3	24.6	25.1	27.3	26.4	21.7	17.3	9.7	1.3	(32)	
(33)		10%	WB	-6.0	-2.2	2.6	7.5	11.0	15.0	17.6	17.2	13.5	7.5	1.2	-4.6	(33)
(34)		MDCB	-2.4	3.2	10.9	19.1	23.7	25.4	26.3	25.2	21.2	15.6	7.5	-0.7	(34)	
(35)	Mean Daily Temperature Range	MDBR	13.9	14.4	13.8	13.3	12.9	12.0	11.4	11.3	11.7	13.1	12.9	13.0	(35)	
(36)		5% DB	MCDBR	16.3	16.7	16.8	16.5	15.3	14.2	13.5	13.1	14.0	15.6	15.4	14.8	(36)
(37)		MCWBR	11.8	10.9	9.3	7.8	6.5	4.6	3.9	4.0	5.1	7.3	8.8	10.4	(37)	
(38)		5% WB	MCDBR	15.7	15.6	15.2	14.6	12.9	10.8	10.7	10.4	10.7	13.7	14.4	14.3	(38)
(39)		MCWBR	11.5	10.3	8.5	7.0	5.8	3.9	3.5	3.7	4.6	6.9	8.5	10.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.353	0.401	0.449	0.472	0.481	0.471	0.452	0.422	0.373	0.328	0.302	(40)	
(41)		taud	2.163	2.039	1.923	1.844	1.846	1.891	1.980	2.015	2.029	2.082	2.136	2.149	(41)	
(42)		Ebn at Noon	834	835	835	821	812	804	808	811	806	802	792	797	(42)	
(43)		Edh at Noon	102	135	171	199	203	195	177	166	152	128	104	95	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.61	3.72	5.03	6.24	7.01	6.94	6.43	6.02	5.09	4.16	2.93	2.38	(44)	
(45)		RadStd	0.21	0.17	0.35	0.29	0.32	0.43	0.35	0.29	0.29	0.16	0.17	0.18	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+63	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+61	N/A

Nomenclature: See separate page